Ranking, Calibration and Cost in Competing-Risks Survival Models

A Common-Protocol Benchmark on 550,000 Mortgages

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The question

Deep competing-risks models are usually compared on how well they **rank** borrowers. A lender provisioning against loss needs the **probability**. We tested whether the ranking advantage holds when both are measured.

Why it matters

A mortgage ends in one of three ways: default, prepayment or maturity. Once one happens the other two cannot, which is what makes this a competing-risks problem.

IFRS 9 requires a probability of default at each future date, and the reserve is computed from it. A model can order borrowers correctly and still get those probabilities wrong.

Data

Freddie Mac Single-Family Loan-Level Dataset, 50,000-loan samples per vintage.

Cohort	Train	Test
Post-crisis	2013–15	2016–18
Crisis	2005–06	2007–09

20.6% vs 95.5%

censoring, counting prepayment as an event versus censoring it

Dirick et al. report 20–85% and 94.56–98.16% across ten European consumer-credit portfolios. Our figures fall inside both ranges on US mortgages from a later decade.

Protocol

Every model returns a cumulative incidence function per cause, so all five are scored on all three metrics.

Models Aalen–Johansen (null), cause-specific Cox, Fine–Gray, random survival forest, DeepHit
Ranking IPCW time-dependent AUC
Calibration integrated calibration index, integrated Brier
Cost fit and predict seconds
Splits vintage-ordered, not random
Budget 20 DeepHit configs, 8 forest, fixed in advance, validated out-of-time
Inference paired bootstrap, $B{=}500$, two-sided, α fixed at 0.05, Bonferroni across three metric families

1. The advantage reverses with regime

AUC at 60 months, default:

	Crisis	Post-crisis
DeepHit (tuned)	**0.8538**	0.7280
Cause-specific Cox	0.8346	**0.7372**
Difference	$+0.0192$	-0.0091

Both are significant under the paired bootstrap. The crisis win is harder to read than it looks: refitting Cox on resamples of its own training vintages moves its AUC by ± 0.0159, against 0.0035 for DeepHit. Here the classical models are the less stable ones.

Calibration behaves the same way in both regimes: DeepHit's calibration index is 0.0040 worse than Cox's in each.

2. DeepHit against a random forest

$p = 0.97$ $p = 0.79$

DeepHit vs tuned random survival forest, AUC, crisis and post-crisis

Neither difference reaches significance, and both are smaller than DeepHit's spread across seeds. On crisis **prepayment** the forest is ahead on both, 0.685 against 0.620 on AUC and 0.154 against 0.236 on calibration.

3. What the loss weight does

DeepHit's loss is α·likelihood + $(1-\alpha)$·ranking. Averaging our 40 tuning configurations by α, validation AUC rises from 0.658 at $\alpha{=}0.1$ to 0.776 at $\alpha{=}1.0$, a strong monotone effect.

We then varied α alone, holding architecture, dropout, learning rate, batch size and bin count fixed, over five seeds and both cohorts:

Cause	Cohort	ΔAUC	seed sd
Default	crisis	$+0.0075$	0.0033
Default	post	$+0.0005$	0.0018
Prepay	crisis	$\mathbf{-0.0673}$	0.0103
Prepay	post	$\mathbf{+0.0386}$	0.0062

Δ is the change from $\alpha{=}0.1$ to $\alpha{=}1$. Three of the four cells exceed the seed standard deviation and they disagree about direction: crisis prepayment loses 0.067 of AUC where post-crisis prepayment gains 0.039.

In the search α varied together with depth, dropout and batch size, so the pooled table measured all of them at once. **We had written the opposite conclusion before we ran the controlled version.**

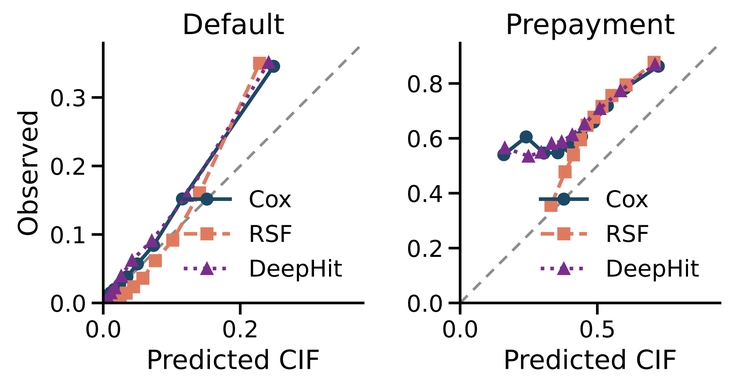
4. Where the calibration gap comes from

Calibration index	Crisis	Post-crisis
Raw	0.0209	0.0318
Recalibrated (train)	0.0164	0.0297
Recalibrated (oracle)	**0.0053**	**0.0082**

We estimated a per-horizon rescaling two ways. Fitted on the training vintages, which is what a lender has at scoring time, it changes very little and in 10 of 16 cells makes calibration worse. Fitted on the test vintages it cuts the error by a median factor of 4.2.

It works when it is estimated on the vintages it will be applied to, and those are the ones a lender does not have yet.

Calibration at 60 months, crisis cohort



Deciles of predicted risk against the Aalen–Johansen estimate. All three models sit above the diagonal, and the gap is far wider for prepayment.

Cost

$\approx 2{,}000\times$

DeepHit (59 s) versus Fine–Gray (0.03 s) per fit, 150,000 loans

The DeepHit search added 14.4 minutes across the two cohorts. The classical models have no search to run.

What we cannot claim

All results come from one family of US agency mortgages. Variability comes from only five bootstrap replicates. Covariates are observed at origination, so we have no time-varying macroeconomic variables. Our recalibration is a marginal rescaling rather than the per-cause procedures of Alberge et al. (2026). With 150,000 test loans most differences reach significance, so we report effect sizes alongside p-values.

DeepHit ranks better in one of two regimes, and is worse calibrated in both.

Code: `github.com/ar-sadraei/rtds-project`

Data: Freddie Mac SFLLD, free on registration